

Rohit Kumar

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PROFESSIONAL SUMMARY

IT undergraduate at VIT Vellore with hands-on **MERN full-stack development** experience through **2 paid freelance client projects**, including institute and salon management platforms. Experienced in role-based applications, real-time communication, administrative dashboards, appointment and enquiry workflows, and database-backed systems. Strong foundation in JavaScript, React, Node.js, Express.js, MongoDB, REST APIs, and AWS. Co-inventor on **3 published Indian patent applications**.

EDUCATION

Vellore Institute of Technology, Vellore, Tamil Nadu
B.Tech in Information Technology | **CGPA: 6.79 / 10.00**

2023 – 2027

TECHNICAL SKILLS

Languages: JavaScript, HTML5, CSS3, Python, Java

Web & Backend: React, JavaScript/JSX, Node.js, Express.js, REST APIs, Socket.IO

Databases: MySQL, MongoDB

Cloud, Deployment & Tools: AWS, Vercel, Render, Git, GitHub, CI/CD, VS Code

Concepts: DBMS, OS, OOP, Computer Networks, Web Technology, Cloud Computing, Software Engineering, Network Security

Soft Skills: Client Communication, Problem Solving, Time Management, Adaptability

PROJECTS

ACE Academy: Institute Management System

Live Demo | Code

React / JavaScript/JSX / Node.js / Express.js / MongoDB / Socket.IO / CSS / JSON

- Engineered and deployed a role-based institute management platform supporting **3 user roles – Admin, Teacher, and Student** – across attendance, timetables, marks, study materials, and student progress tracking, backed by **RESTful APIs** built with Node.js and Express.js.
- Integrated administrative account provisioning and credential management with **real-time student-teacher communication using Socket.IO**; deployed on Vercel and Render with Git-based **CI/CD**.

Platinum Unisex Salon: Management Website

Live Demo | Code

JavaScript / MongoDB / CSS / JSON

- Delivered a production client website integrating **online appointment booking, customer enquiries, service information, locations, and operating hours**.
- Created an admin dashboard for centralized enquiry and booking management, including appointment-status updates after service completion.

Taxi Route Anomaly & Overcharging Detection

Code

Python / PySpark / Apache Spark / Pandas / Flask / Scikit-learn

- Processed a **32 GB NYC Yellow Taxi 2023 dataset** using PySpark and Apache Spark, performing data cleaning, feature engineering, and route-level statistical analysis.
- Derived fare-per-mile, fare-per-minute, and trip-duration baselines and applied $\mu \pm 2\sigma$ **anomaly detection**, with a Flask dashboard for route-level anomaly analysis.

Sentinel Entropy IDS: Zero-Day Traffic Anomaly Detection

Code

Python / Flask / Pandas / Scapy / Shannon Entropy / HTML/CSS

- Engineered an anomaly-based IDS supporting **CSV, PCAP, and PCAPNG** traffic analysis using Scapy and Shannon entropy across IPs, ports, protocols, and packet-size distributions.
- Established statistical traffic baselines and delivered a Flask dashboard presenting anomaly windows, risk scores, protocol analysis, top hosts, and watched ports.

PATENTS

Co-inventor on 3 published Indian patent applications:

- System and Method for Time-Aware Retrieval-Augmented Predictive Behavioral Risk Scoring in Cognitive Care Settings** – IN202641031052.
- System and Method for Engagement-Adaptive Multimodal Retrieval-Augmented Personalized Reminiscence Interaction**.
- ϵ -bound-verified fuzzy-rule surrogate modeling for interpretable LLM compression** – IN202641043575.

PROFESSIONAL EXPERIENCE

Freelance Full-Stack Developer

Mar 2026 – Jun 2026

Self-Employed, Vellore, Tamil Nadu, India

- Delivered **2 paid client projects** independently, handling requirements, full-stack implementation, MongoDB integration, user workflows, administrative functionality, and production deployment.

CERTIFICATIONS

- Advanced Generative AI** – IBM Career Education Program / IBM Developer Skills Network, 2026.
- Diploma in Python (PYV101)** – Independent Skill Development Mission, **94%, Grade A+**, 2023.

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